Inherited F4: eager rewrite of kept rows

2026-10-11. Change: `src/cryptanalysis/inherited_f4.rs` (PR branch `inherited-f4-eager-rewrite`). Control: `KIC_F4_EAGER=0`, same binary. Run report: `reports/run-20261011T1657Z/report.pdf`.

Question

Where does a specialisation of an inherited Macaulay basis spend its time once the decomposition tree is fixed, and can it be cut without changing any row space, verdict or tree counter?

Mechanism

Before the change, a specialisation $v := c$ kept the rows whose pivot avoids v by reference, in their parent's layout, and rewrote a row into the child's layout only when a displaced row's re-reduction hit it as a pivot (lazy rows). A row last current k layouts back needed the composed map of k steps and a `pext`/`pdep` plan for that pair of layouts, built word by word on first use, plus a staging pass, a copy and one allocation per rewrite.

Callgrind on one n = 23, dim 8 target of the F6-IC arm (this host, the parent build, not committed as a record) counted 260 521 lazy rewrites at 622 specialisations, about 70 % of the rows of a basis at every node, with the lazy rewrite, the plan construction and the allocator together near 40 % of instructions. Since most rows are rewritten at every node anyway, the change rewrites every kept row at once, through one flat plan of the step's own map, into a single buffer shared by the node's rows (`StepPlan`, `ReducedBasis::rewrite_kept`); displaced rows use the same plan. Rows from an older layout, which occur only under a lazy ancestor, still take the lazy path.

The row space of every basis is unchanged: each row's image is the same vector, only computed earlier. The specialisation word count (`word_xors`) rises because every kept row's rewrite is now charged.

Result

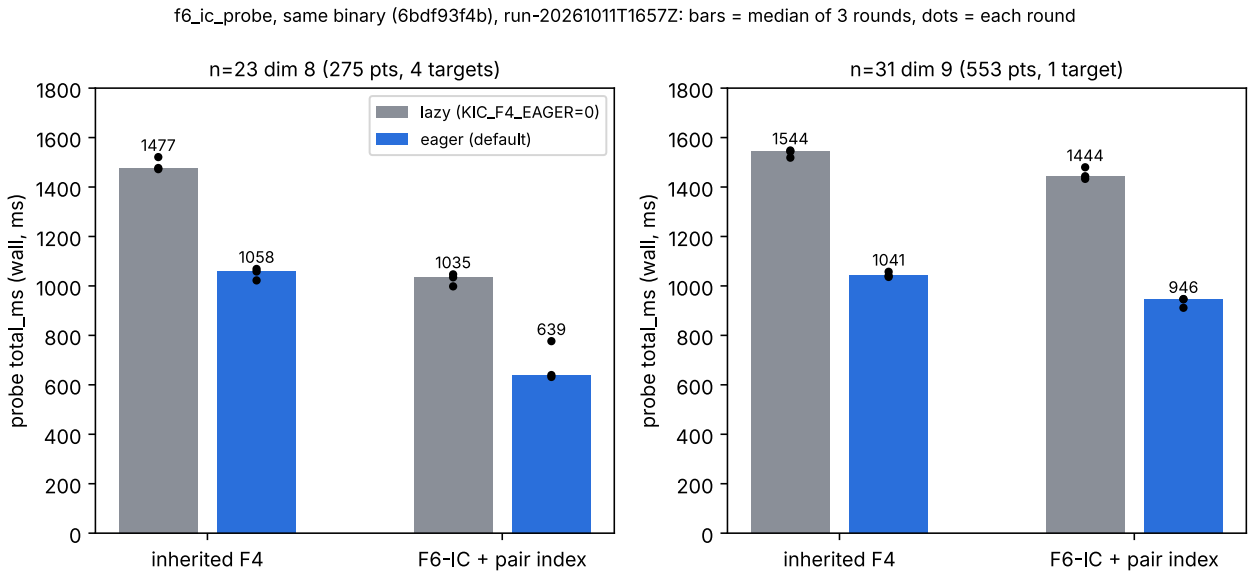


Figure 1: `f6_ic_probe` totals, run-20261011T1657Z, 3 interleaved rounds per mode (dots), bars at the median.

cell	arm	lazy median ms	eager median ms	ratio lazy/eager	counters
$K_0/2^{23}$ dim 8, 4 targets	inherited F4	1477.0	1058.4	1.40	identical
$K_0/2^{23}$ dim 8, 4 targets	F6-IC + index	1034.8	639.5	1.62	identical
$K_0/2^{31}$ dim 9, 1 target	inherited F4	1543.5	1041.3	1.48	identical
$K_0/2^{31}$ dim 9, 1 target	F6-IC + index	1443.6	946.1	1.53	identical

Medians measured (3 rounds each); ratios derived as lazy median / eager median. In every round each eager value is below every lazy value of the same cell and arm. "Identical": reductions, splits, gate counters and every target's verdict.

Scope and limits

- One host, isolation L0 (shared cloud container, no pinning), three rounds per mode; no bootstrap interval. The ratios are wall-time observations of the probe on this host, not an ecbench comparison.
- Solver-internal only: this changes the constant of the Gröbner track, not its exponent. The decomposition search still enumerates about $|F|^{m-2}$ partial assignments, and nothing here compares index calculus with rho.
- `cargo test --release --lib`: the `inherited_f4::` (both modes), `koblitz_groebner::` and `f6_ic` tests pass. The full suite's 12 failures are in modules that do not reach this code (curve registry parsing, prime BSGS, BIKE, an ecbench record test, n71 construction).

Graphs checked and left unchanged

`docs/ic/progress-timeline.json`, `docs/index-calculus-scoreboard.html`, `docs/performance-gains/`: they plot IC-versus-rho accounting from ecbench sessions and the specialisation study of `performance-gains`; a probe-level solver speedup without an ecbench session changes none of their values or conclusions.